# Determining the Opportunities and Challenges for Implementing Smart Multi-Channel Digital Sub-Metering Systems for Shared Residential Units in Barangay San Jose, Puerto Princesa City

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***Abstract* — This study assesses the socio-economic and operational viability of implementing Smart Multi-Channel Digital Sub-Metering Systems in shared residential units in Barangay San Jose, Puerto Princesa City. Currently, localized micro-rental properties rely predominantly on analog sub-meters or manual equal bill splitting, resulting in widespread administrative inefficiency and a high tenant dispute rate of 55.9%. Utilizing a quantitative descriptive-survey research design, data was gathered through purposive sampling from 12 boarding house owners and 34 tenants to evaluate perceptions on fairness, transparency, operational efficiency, and economic viability. The findings reveal a highly positive reception toward digital integration. Despite minor concerns regarding local Wi-Fi stability and data privacy, both stakeholders strongly favor automated, real-time consumption tracking. Economically, the system is highly viable; 91.7% of landlords are willing to invest in the technology upon realizing the estimated ₱3,300 hardware cost is comparable to traditional setups. Concurrently, 85.3% of tenants expressed a willingness to pay a monthly premium (predominantly ₱50 to ₱200) to ensure accurate, dispute-free billing. The study concludes that smart digital sub-metering is socio-economically sustainable and highly demanded in the locale. It is recommended that future implementations utilize "offline-first" microcontroller architectures to mitigate internet interruptions and deploy aggregate data dashboards to secure tenant privacy.**

**Keywords— Digital Sub-Metering, Internet of Things (IoT), Analog Sub-Metering, Boarding Houses, Technology Acceptance.**

## I. Introduction

### *A. Background of the Study*

The rapid urbanization of Puerto Princesa City, Palawan, particularly in commercial and educational centers like Barangay San Jose, has led to a significant increase in the demand for shared residential units, such as boarding houses and apartments. In these shared living arrangements, energy management and billing present a constant challenge. Currently, most landlords rely on traditional analog sub-meters billing or single main sub-meter for the entire property where bills are divided manually among tenants. While simple, these methods often lead to inaccuracies, a lack of transparency, and frequent disputes between landlords and tenants.

Smart multi-channel digital sub-metering systems offer a modern alternative. These Internet of Things (IoT)-based systems provide real-time data monitoring and automated bill splitting. While the technical viability of microcontrollers is well-documented, the actual adoption of these systems depends heavily on socio-economic and operational factors.

### *B. Statement of the Problem*

This study aims to determine the opportunities and challenges of implementing smart multi-channel digital sub-metering systems for shared residential units in Barangay San Jose, Puerto Princesa City. Specifically, it seeks to answer:

1. How do boarding house owners and tenants in Barangay San Jose perceive the fairness, transparency, and administrative efficiency of smart digital sub-metering systems?
2. What is the willingness to pay (WTP) among tenants for the integration of smart sub-metering features, and how do landlords evaluate its cost-efficiency?
3. What are the primary opportunities and challenges that will drive or hinder the adoption of this technology?

### *C. Objectives of the Study*

The main objective is to assess the opportunities and challenges of deploying these systems. Specifically, it aims to:

1. Evaluate social and operational impacts regarding fairness, transparency, and administrative benefits.
2. Determine economic viability by measuring tenants' WTP and landlords' cost-efficiency perceptions.
3. Identify core opportunities and challenges to adoption.

### *D. Significance of the Study*

The findings will benefit boarding house owners by providing data-driven insights on financial and administrative returns. It will advocate for fair billing for tenants, guide IoT technology providers in pricing localized solutions, and serve as a socio-technical reference for future Computer Engineering researchers.

### E. *Scope and Delimitation*

This study is strictly confined to assessing the socio-economic and operational opportunities and challenges of smart digital sub-metering. The geographic scope is limited to boarding houses in Barangay San Jose, Puerto Princesa City. It does not involve physical prototyping, hardware fabrication, or software programming. To ground the economic and willingness-to-pay assessments, a conceptual system architecture and estimated bill of materials are presented in Section II.

## II. Conceptual System Architecture And Cost Estimation

### A. *System Architecture and How it Works*

The proposed system operates on an Internet of Things (IoT) framework designed specifically for multi-room shared residential units. Instead of placing a separate, expensive utility meter in each room, a single centralized microcontroller node is installed near the boarding house's main distribution panel.

1. *Sensing Layer:* The system utilizes non-invasive split-core current transformers and voltage sensors (such as the PZEM-004T module) clipped onto the live wires of each individual room. These sensors accurately measure the real-time voltage, current, active power, and energy consumption of up to four or more separate rooms simultaneously.
2. *Processing Layer:* A low-cost, Wi-Fi-enabled microcontroller (such as an ESP32) acts as the brain of the system. It continuously reads the data from the sensors, calculates the power usage per channel, and processes the localized logic.
3. *Application and Cloud Layer:* The microcontroller transmits the calculated energy data via the local Wi-Fi network to a secure cloud database. Landlords and tenants can then access this data through a mobile application or web dashboard, which automatically converts the kilowatt-hour (kWh) readings into monetary values based on the current local utility rates.

### B. *Core Features and Functions*

The system is designed to replace traditional manual computation with the following features:

- *Multi-Channel Monitoring:* The ability to monitor several independent rooms from a single microcontroller node, drastically reducing hardware costs.
- *Automated Bill Splitting:* The software automatically computes the exact peso amount owed by each room based on their distinct consumption, eliminating the need for manual math and reducing human error.
- *Real-Time Transparency:* Tenants can view their daily consumption on their smartphones, preventing "bill shock" at the end of the month.
- *Remote Cut-off (Optional):* Integrated relay modules allow the landlord to remotely disable power to specific rooms in cases of severe non-payment or safety emergencies.

### C. *Estimated Fabrication Cost*

To accurately measure the landlords' return on investment (ROI) and the tenants' Willingness to Pay (WTP), a theoretical Bill of Materials (BOM) for a standard 4-room boarding house configuration is estimated as follows:

- Main Microcontroller (e.g., ESP32): ₱300.00
- Power/Energy Sensors (4x PZEM-004T): ₱2,000.00
- 4-Channel Relay Module: ₱250.00
- Power Supply and PCB layout: ₱350.00
- Enclosure, Wiring, and Connectors: ₱400.00
- *Total Estimated Hardware Cost:* ₱3,300.00

## III. Literature Review

### A. *Technological Readiness of IoT Energy Monitoring*

The transition toward modernized energy management relies heavily on IoT technologies. Recent implementations prove that low-cost microcontrollers, such as the ESP32 paired with sensors, can accurately monitor real-time energy consumption [1], [2]. These systems provide consumers with mobile dashboards for remote monitoring [3]. To address data privacy, researchers have proposed decentralized storage solutions, ensuring user consumption data remains secure [5].

### B. *Billing Innovations and Dispute Resolution*

In shared units, estimated billing leads to tension. Digital sub-metering directly mitigates these issues. The implementation of smart prepaid systems allows users to pay-as-they-go, automatically restricting energy usage when credits are exhausted, effectively eliminating billing disputes [6].

### C. *Behavioral Barriers and Policy Challenges*

Despite proven reliability, smart meter adoption faces socio-economic hurdles. At a macro level, the European transition toward sub-metering is delayed by legal frameworks regarding data ownership [7]. At the consumer level, behavioral resistance is a primary barrier. A case study in Kashmir revealed massive public protests due to dynamic pricing clashing with the region's harsh winters [8]. This "status quo bias" demonstrates that without clear economic incentives, rollouts face severe pushback [8].

### D. *Local Context in the Philippines*

The feasibility of localized smart metering has been validated in the Philippines. A recent study successfully fabricated a smart energy meter using an ESP32 tailored for Philippine voltage standards, achieving high accuracy and Wi-Fi stability, proving DIY IoT solutions are highly viable for local settings [9].

### E. *Synthesis*

The literature establishes a clear consensus: the hardware required for digital sub-metering is mature and cost-effective [1]–[5]. However, the true bottlenecks are human factors: policy, privacy, and public perception [7], [8]. This study bridges the gap by focusing purely on the localized operational impacts, social perceptions, and financial willingness of landlords and tenants in Barangay San Jose.

## IV. Methodology

### A. Research Design

This study utilizes a quantitative descriptive-survey research design. This method is highly appropriate for assessing the current conditions, practices, and socio-economic perceptions of a specific population. Since the objective is to determine the opportunities, challenges, and willingness to pay (WTP) regarding smart digital sub-metering without manipulating variables or deploying physical prototypes, the descriptive approach allows for the systematic and objective collection of data from the target demographic.

### B. Research Locale

The study will be conducted exclusively in Barangay San Jose, Puerto Princesa City, Palawan. Barangay San Jose is a highly urbanized and commercialized district that hosts several educational institutions (including Fullbright College) and major business establishments. This concentration makes it a primary hub for boarding houses and shared residential apartments catering to students and transient workers, providing an ideal and highly relevant setting for analyzing localized micro-rental energy economics.

### C. Respondents of the Study and Sampling Technique

The respondents of this study are categorized into two distinct groups: boarding house owners/landlords and their respective tenants.

To select the respondents, the researchers will use purposive sampling to ensure representation from the two primary billing setups currently existing in the locale:

1. Boarding houses that utilize *individual analog sub-meters* for each room.
2. Boarding houses that rely on a *single main sub-meter* for the entire property (where bills are divided manually among tenants).

The criteria for selection require that the landlord operates a shared residential unit within Barangay San Jose, and the tenant currently resides in one of these setups and pays for a share of the electricity. The target sample size is 10 to 15 boarding house owners and 30 to 50 tenants to ensure a comprehensive representation of both the management and consumer perspectives across different existing billing infrastructures.

### D. Research Instrument

The primary data gathering tool is a structured, researcher-made survey questionnaire tailored separately for landlords and tenants. The instrument is divided into the following sections:

- *Part I: Demographic and Current Billing Profile.* Collects basic information and categorizes the respondent based on their current setup (individual analog sub-meter vs. single main meter with manual splitting).
- *Part II: Social and Operational Perceptions.* Utilizes a 4-point Likert scale (Strongly Agree to Strongly Disagree) to measure perceptions regarding fairness, transparency, privacy concerns, and the anticipated efficiency of dispute resolution if upgraded to a smart digital system.
- *Part III: Economic Viability and Willingness to Pay.* A mix of multiple-choice and scaled questions to determine the specific monetary amount tenants are willing to add to their monthly rent for smart features, and how landlords perceive the return on investment (ROI).

### E. Data Gathering Procedure

The researchers will deploy the survey digitally using Google Forms. Links and scannable QR codes directing to the online questionnaire will be distributed to the selected boarding house owners and tenants through direct messaging platforms or actual visits to the residential units. The initial section of the Google Form will feature an electronic informed consent agreement, explicitly assuring all respondents that their identities and specific financial data will remain strictly confidential and will be utilized solely for academic purposes. The researchers will actively monitor the real-time data collection via the Google Forms dashboard and conduct digital or in-person follow-ups to ensure the target sample size is successfully reached.

### F. Statistical Treatment of Data

The data gathered from the survey will be tallied, tabulated, and analyzed using the following statistical tools:

1. *Frequency and Percentage Distribution:* This will be used to analyze the demographic profile of the respondents, the prevalence of the two different current billing methods, and their categorized willingness to pay (WTP) amounts.
   *Formula:* $P = \frac{f}{N} \times 100$
   (Where *P* is the percentage, *f* is the frequency, and *N* is the total number of respondents).
2. *Weighted Mean:* This will be utilized to quantify the responses from the Likert-scale sections regarding social perceptions, fairness, and operational challenges. By comparing the weighted means of tenants using analog sub-meters versus those using a single main meter, the researchers can determine which group perceives a higher need for a smart digital upgrade.
   *Formula:* $WM = \frac{\Sigma fx}{N}$
   (Where *WM* is the weighted mean, *f* is the frequency of responses, *x* is the weight of each option, and *N* is the total number of respondents).

## V. Results and Discussion

This chapter presents the data gathered from the boarding house owners and tenants in Barangay San Jose, Puerto Princesa City. The data is organized, tabulated, analyzed, and interpreted following the sequence of the specific objectives posed in Chapter I.

### A. Demographic and Current Billing Profile

To establish the baseline context of the micro-rental market in Barangay San Jose, the researchers first evaluated the current property management and billing systems utilized by the respondents.

Among the 12 landlord respondents, the majority manage small-scale boarding houses, with 58.3% managing 2 to 5 rooms, 25.0% managing 6 to 10 rooms, and 16.7% managing large-scale dormitories (more than 10 rooms). When asked how they currently distribute the monthly PALECO bill, a significant 66.7% stated they simply divide the total PALECO bill equally among their tenants, while only 33.3% rely on traditional analog sub-meters for each room. The majority of landlords (58.3%) spend less than 1 hour per month calculating these bills, while 41.7% spend between 1 to 3 hours.

This landscape is reflected in the 34 tenant respondents. A majority of tenants (52.9%) reported paying through manual equal bill splitting, while 47.1% pay via an analog sub-meter (see Fig. 1). More than half of the tenants (52.9%) have lived in their units for more than a year. Notably, 55.9% of the tenant respondents have experienced a dispute or felt they were overcharged for their electricity share under these current manual setups. This high dispute rate underpins the necessity of introducing a more transparent, data-driven system in the locale.

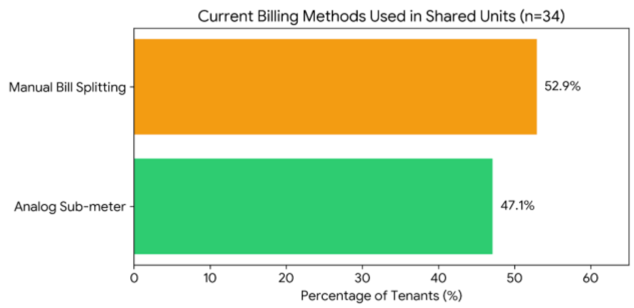


**Fig. 1.** Current electricity billing methods experienced by tenant respondents in Barangay San Jose.

*B. Social and Operational Perceptions*

The second phase of the survey assessed how stakeholders perceive the fairness, transparency, privacy, and operational impacts of upgrading to a smart multi-channel digital sub-metering system. The data was quantified using a 4-point Likert Scale, interpreted as follows: 3.25–4.00 (Strongly Agree), 2.50–3.24 (Agree), 1.75–2.49 (Disagree), and 1.00–1.74 (Strongly Disagree).

1. *Tenant Perceptions on Fairness and Transparency*: The tenants showed a highly positive reception toward the technology. They Strongly Agreed (WM = 3.29) that having real-time access to their consumption via a mobile app would help them trust the billing process more. They Agreed (WM = 3.21) that the system would eliminate arguments with their landlords, and Agreed (WM = 3.09) that it would make the computation fairer.

   When it comes to privacy, tenants Agreed (WM = 3.15) that they are comfortable with a digital system tracking their real-time usage. However, they also expressed some hesitation, Agreeing (WM = 3.03) that they have concerns the system might invade their privacy by showing exactly when they use certain appliances. This indicates that while transparency is highly desired, data privacy features must be carefully implemented (see Fig. 2).

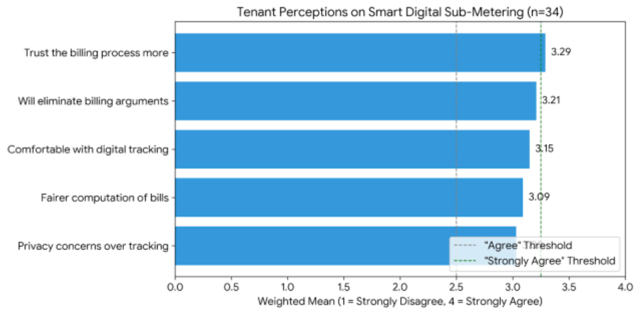


**Fig. 2.** Weighted mean distribution of tenant perceptions regarding digital sub-metering.

2. *Landlord Perceptions on Operational Efficiency*: From an operational standpoint, the landlords uniformly Strongly Agreed (WM = 3.25) on three critical factors: that a smart system would significantly reduce the time they spend manually computing bills, that automated billing would reduce tenant complaints, and that the initial cost of purchasing IoT sub-meters is a major barrier.

   Furthermore, landlords Strongly Agreed (WM = 3.25) that they are concerned about the local internet (Wi-Fi) stability in Barangay San Jose affecting the cloud-based system's performance. Despite these concerns, they still Agreed (WM = 3.00) that they feel confident in their ability to manage and operate an app-based dashboard.

*C. Economic Viability and Willingness to Pay (WTP)*

The final objective was to determine the financial feasibility of deploying the system, specifically measuring the tenants' willingness to pay (WTP) and the landlords' willingness to invest.

1. *Landlord Willingness to Invest*: During the survey, landlords were informed that the estimated cost of a 4-room smart sub-metering setup is approximately ₱3,300, which is highly competitive with purchasing four traditional analog meters. With this financial context, an overwhelming 91.7% of landlord respondents stated they are willing to invest in the digital upgrade. Interestingly, when asked how they prefer to recover this cost, exactly half (50.0%) stated they are willing to absorb the cost entirely as a property upgrade to attract more tenants. Meanwhile, 33.3% prefer slightly increasing the base rent, and 16.7% prefer adding a separate tech/administrative fee.

2. *Tenant Willingness to Pay (WTP):* The financial willingness of the tenants validates the landlords' investment confidence. When asked if they would be willing to pay a slightly higher monthly rent to cover the cost of a system that guarantees transparent

billing, a massive 85.3% of the tenants answered "Yes." Among the tenants willing to pay, the maximum additional amount they are willing to absorb per month is distributed as follows (see Fig. 3):

₱50 - ₱100: 40.0%

₱101 - ₱200: 30.0%

₱201 - ₱300: 23.3%

More than ₱300: 6.7%

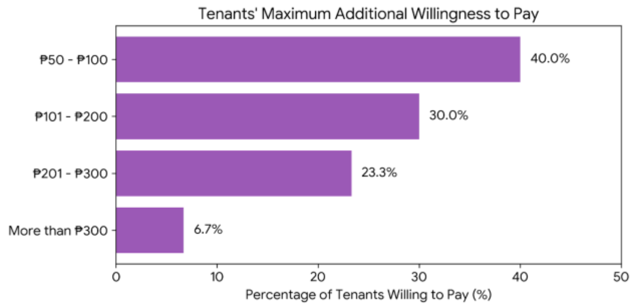


**Fig. 3.** Maximum additional monthly amount tenants are willing to pay for smart billing integration.

### *D. Synthesis of Opportunities and Challenges*

Based on the gathered empirical data, the deployment of smart digital sub-metering in Barangay San Jose presents distinct opportunities and challenges:

Opportunities: The market readiness is exceptionally high. With a 55.9% dispute rate under current manual billing, both tenants and landlords are actively seeking automated transparency. The financial viability is fully validated: 91.7% of landlords are willing to invest, and 85.3% of tenants are willing to pay an additional premium (mostly between ₱50 and ₱200) for the technology.

Challenges: The primary barriers hindering immediate adoption are technical and infrastructural rather than financial. Landlords are highly concerned about Wi-Fi reliability disrupting automated billing, while tenants maintain moderate privacy concerns regarding appliance-level tracking. Overcoming these challenges will require robust offline-fallback features in the microcontrollers and clear data-privacy agreements between property owners and renters.

## VI. Summary Of Findings, Conclusions, And Recommendations

This chapter presents a synthesis of the data gathered from the survey, the definitive conclusions drawn from the analysis, and actionable recommendations for stakeholders and future researchers.

### *A. Summary of Findings*

The study assessed the socio-economic and operational viability of implementing smart multi-channel digital sub-metering systems in Barangay San Jose, yielding the following key findings:

1. *Current Billing Landscape*: The majority of landlords (66.7%) divide the total PALECO bill equally, while only 33.3% use analog sub-meters. This manual approach has led to a high dispute rate, with 55.9% of tenants reporting they have experienced conflicts or felt overcharged for electricity.
2. *Social and Operational Perceptions*: Tenants exhibited a strong desire for transparency, strongly agreeing (WM = 3.29) that mobile app access would build trust in the billing process. However, they expressed moderate privacy concerns (WM = 3.03) regarding appliance-level tracking. Landlords strongly agreed (WM = 3.25) that automation would save time and reduce complaints, but they also strongly agreed (WM = 3.25) that the local Wi-Fi stability in Palawan could disrupt the system.
3. *Economic Viability*: The financial readiness for adoption is exceptionally high. Upon learning that the estimated ₱3,300 hardware cost is comparable to traditional setups, 91.7% of landlords expressed willingness to invest, with 50.0% willing to absorb the cost as a property upgrade. Concurrently, 85.3% of tenants are willing to pay a premium for the technology, predominantly between the ₱50 to ₱200 monthly range.

### *B. Conclusions*

Based on the findings, the study concludes the following regarding the specific research questions:

1. *On Perceptions of Fairness, Transparency, and Efficiency*: Both landlords and tenants perceive smart digital sub-metering as a highly effective tool for mitigating the current 55.9% dispute rate. While landlords see clear administrative benefits in time-saving and conflict resolution, tenants view the system as a necessary mechanism for fair and transparent billing, provided their privacy is respected.
2. *On Economic Viability and Willingness to Pay*: The economic ecosystem in Barangay San Jose is fully primed for IoT smart metering. The financial gap between landlord investment and tenant willingness to pay (WTP) is virtually non-existent. Tenants are willing to absorb micro-increases in rent (₱50-₱200), which perfectly aligns with the landlords' preferred ROI strategies, proving the technology is financially sustainable in this micro-rental market.
3. *On the Primary Opportunities and Challenges*: The primary opportunity lies in the overwhelming market demand and financial feasibility; the barrier to entry is no longer cost. The primary challenges are infrastructural and behavioral—specifically, the landlords' valid fears regarding the island's Wi-Fi stability, and the tenants' reservations about invasive data tracking.

## C. Recommendations

In light of the conclusions, the researchers recommend the following:

1. *For Boarding House Owners and Landlords*: It is recommended to transition toward digital sub-metering systems as a property upgrade to attract tenants who value transparent billing. Landlords can initially offset the ₱3,300 hardware cost by introducing a nominal ₱50 monthly administrative fee, which the data proves is highly acceptable to renters.

2. *For Technology Providers and System Integrators*: When developing these systems for the Palawan market, developers must design "Offline-First" architectures. Microcontrollers must be capable of logging kilowatt-hour data locally on an SD card during Wi-Fi outages and syncing to the cloud only when the connection is restored. Furthermore, to address tenant privacy concerns, mobile dashboards should display aggregated daily consumption rather than real-time, appliance-specific profiling.

3. *For Future Computer Engineering Researchers*: Since this study has validated the socio-economic and financial viability of the system in Barangay San Jose, it is highly recommended that future researchers proceed to the next phase: the physical fabrication, coding, and deployment of the ESP32 and PZEM-004T multi-channel prototype in an actual boarding house setting to test real-world network latency and hardware endurance.


# VII. Acknowledgment

The completion of this research would not have been possible without the guidance, support, and encouragement of numerous individuals and institutions. The researchers would like to extend their deepest gratitude and sincere appreciation to the following:

To Ms. Nikka Allen Santos, our esteemed research adviser and instructor, for her unwavering patience, continuous guidance, and brilliant insights. Her constructive criticisms and technical expertise were instrumental in shaping this study from a mere concept into a comprehensive manuscript.

To Fullbright College, for providing a conducive environment for academic excellence and for equipping us with the foundational knowledge required to undertake this socio-technical assessment.

To the Boarding House Owners, Landlords, and Tenants of Barangay San Jose, who served as the primary respondents of this study. Their willingness to share their time, honest opinions, and personal billing experiences formed the very lifeblood of this research.

To the families, parents, and loved ones of the researchers, for their endless moral encouragement, understanding, and invaluable financial support that sustained the group throughout this endeavor.

Above all, to the Almighty God, for granting the researchers the wisdom, strength, good health, and perseverance to successfully overcome the challenges and complete this academic milestone.

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